

# MacSync-WX1-BO Battery-Operated RS485 & Analog to Wi-Fi Data Logger Datasheet

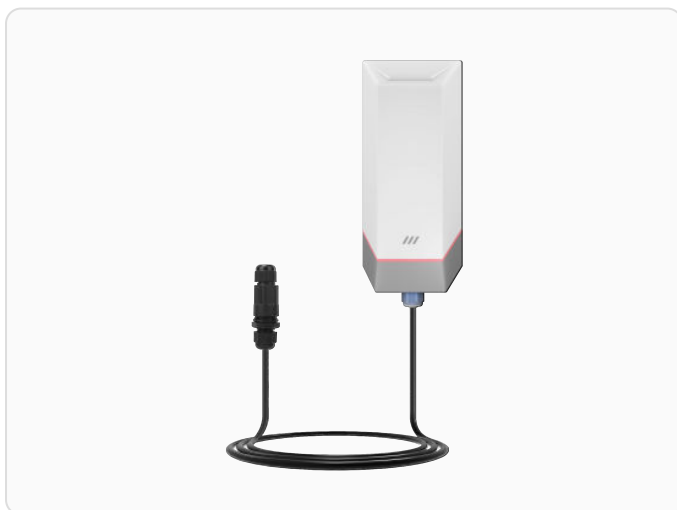
## Product Overview

**MacSync-WX1-BO** is a high-reliability, **battery-operated RS485 & Analog to Wi-Fi data logger**, designed for industrial and on-premise monitoring applications where wired power is unavailable but Wi-Fi connectivity is present. It combines the proven MacSync data-logging hardware with an optimized low-power Wi-Fi communication architecture.

The device interfaces seamlessly with **MODBUS RTU sensors, energy meters, PLCs, transmitters, and industrial instruments**, supporting **RS485, 4–20 mA, 0–10 V, and digital inputs** on a single platform. Collected field data is transmitted securely to **local servers, on-premise dashboards, or cloud platforms** using standard IP protocols such as **MQTT, HTTP/HTTPS**, without the need for gateways.

Powered by a high-capacity **19 Ah industrial-grade Li-SOCl<sub>2</sub> battery**, MacSync-WX1-BO is engineered for extended field operation using intelligent sleep-wake cycles, event-based reporting, and burst data transmission. **Onboard data buffering and store-and-forward mechanisms** ensure zero data loss during temporary Wi-Fi or server outages.

Housed in a rugged IP-rated enclosure, MacSync-WX1-BO is ideal for **factories, buildings, utilities, cold storage facilities, and industrial infrastructure** where battery operation, low latency, and direct Wi-Fi connectivity are required. It enables reliable industrial data acquisition with simplified deployment and full IT/OT integration.





## Key Features

### RS485 (MODBUS RTU) Interface

Seamlessly connects with PLCs, energy meters, RTUs, and industrial controllers using MODBUS RTU.

### Battery-Operated Design

Powered by a 19 Ah industrial-grade Li-SOCl<sub>2</sub> battery for long-term operation without external power.

### Multi-Channel Data Acquisition

Enables simultaneous monitoring of multiple process parameters from RS485 and analog inputs.

### Standard IP Protocol Support

Supports MQTT, HTTP, and HTTPS for seamless IT/OT system integration.

### Event, Threshold & Time-Based Logging

Flexible data logging modes to optimize battery life and network usage.

### Rugged Industrial Enclosure

IP-rated enclosure suitable for factories, buildings, utilities, and harsh environments.

### 4–20 mA, 0–10 V & Digital Inputs

Supports industry-standard analog sensors, transmitters, and digital field signals.

### Low-Power Wi-Fi Communication

Optimized Wi-Fi operation using intelligent sleep-wake cycles and burst data transmission.

### Direct Wi-Fi Connectivity

Communicates directly with local servers and cloud platforms without requiring gateways.

### Onboard Data Buffering

Store-and-forward mechanism prevents data loss during temporary Wi-Fi or server outages.

### Local Configuration via BLE

Easy commissioning and diagnostics using the Macnman Maya mobile application.

### Designed for On-Premise Deployments

Ideal for applications requiring low latency, local data ownership, and battery operation.

# Suited Applications

## Industrial Automation & SCADA

Logs data from PLCs, RTUs, and industrial controllers over RS485 and publishes it to on-premise SCADA systems via Wi-Fi.

## Process & Instrumentation Monitoring

Captures pressure, flow, level, and temperature data from 4–20 mA and 0–10 V industrial transmitters.

## Cold Storage & HVAC Systems

Monitors temperature, humidity, compressors, and HVAC equipment with low-latency Wi-Fi reporting.

## Warehouses & Indoor Infrastructure

Suitable for battery-operated monitoring of power, environment, and equipment across large indoor facilities.

## Energy & Utility Metering

Connects RS485 energy, water, and gas meters for local monitoring and analytics using existing Wi-Fi networks.

## Factories & Manufacturing Plants

Enables battery-powered machine data logging in areas without wired power but with internal Wi-Fi coverage.

## Building Automation & BMS

Integrates meters, controllers, and sensors with BMS / IBMS platforms without requiring gateways.

## On-Premise Industrial IoT Deployments

Ideal for enterprises requiring local data ownership, low latency, Wi-Fi connectivity, and battery operation.

# Technical Specifications

## General Specifications

|                   |                              |
|-------------------|------------------------------|
| Part Order Number | MacSync-WX1-BO               |
| Product Brand     | Macnman Technologies Pvt Ltd |

## Interfaces Available

|                         |                  |
|-------------------------|------------------|
| RS485 Interface         | 1 (Switchable)   |
| 4–20 mA Analog Channels | x 2 (Switchable) |
| 0–10 V Analog Channels  | x 2 (Switchable) |
| Digital Inputs          | x 2 (Switchable) |
| ADC Resolution          | 12-bit           |

**Note**

The device use same channels for the RS485 & Analog inputs, either can be used at a time.

## Wireless Communication

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| Wi-Fi Standards      | IEEE 802.11 b/g/n/ac                                       |
| Frequency Bands      | 2.4 GHz & 5 GHz (Dual-Band)                                |
| Operating Modes      | Station Mode (Client)<br>Access Point Mode (Configuration) |
| Network Protocols    | MQTT, HTTP, HTTPS                                          |
| Security             | WPA2-PSK / WPA3<br>TLS / SSL Encrypted Communication       |
| IP Configuration     | DHCP / Static IP                                           |
| Typical Indoor Range | Up to 100 m (environment dependent)                        |

## LED Status

|                     |                                                    |
|---------------------|----------------------------------------------------|
| LED Type            | 1 Internal Multi-Color LED                         |
| Solid Blue          | Device is connected with Maya App                  |
| Flashing Blue       | OTA (Over-the-Air) update in progress              |
| Slow Blinking Green | Attempting to connect to the network               |
| Single Red Blink    | A confirmed parameter of MacSync is being modified |

## Device Configuration

|                      |                                               |
|----------------------|-----------------------------------------------|
| Configuration        | Via Macnman Maya Android App                  |
| Uplink Modes         | Time Based / Threshold Based / Sampling Based |
| Stored Records*      | 30,000 Records                                |
| Remote Configuration | JSON Format Data - from server                |

## Supported Cloud & Platforms

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| Deployment Models   | On-Premise Servers<br>Private Cloud Deployments<br>Public Cloud Platforms  |
| IoT Cloud Platforms | AWS IoT Core<br>Microsoft Azure IoT Hub<br>Google Cloud (via MQTT / HTTPS) |

|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
|                     | Custom IoT Platforms                                                    |
| Integration Methods | MQTT Brokers<br>REST / HTTPS APIs<br>Webhooks                           |
| Enterprise Systems  | SCADA Systems<br>BMS / IBMS Platforms<br>Data Lakes & Analytics Engines |

## Physical Characteristics

|                       |                                       |
|-----------------------|---------------------------------------|
| Power Supply          | 9-36 V DC                             |
| Operating Temperature | -40°C to +80°C                        |
| Humidity Range        | 0% to 100% RH (non-condensing)        |
| Ingress Protection    | IP65 standard (IP67 optional variant) |
| Housing Material      | Glass field Nylon                     |
| Color                 | Matte White                           |
| Weight                | ~110g                                 |
| Mounting Options      | Pole Mounting / Wall Mounting         |

## Electrical Characteristics

|                     |                                                      |
|---------------------|------------------------------------------------------|
| Input Voltage Range | 9-36 V DC (External Supply)                          |
| Power Consumption   | ~100 µA in sleep mode<br>~150 mA during transmission |
| Input Impedance     | >10 MΩ for voltage inputs                            |

## Communication Options

|                  |                                                      |
|------------------|------------------------------------------------------|
| BLE Connectivity | For local configuration and diagnostics via Maya App |
| Cloud Protocols  | MQTT / HTTPS                                         |
| Serial Interface | RS485 Modbus RTU support (optional)                  |

## Certifications & Compliance

|     |           |
|-----|-----------|
| WPC | Compliant |
|-----|-----------|

|      |                            |
|------|----------------------------|
| RoHS | Conforms to RoHS directive |
|------|----------------------------|

## Security & Encryption

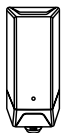
|                     |                                    |
|---------------------|------------------------------------|
| Data Encryption     | AES-256 end-to-end encryption      |
| Authentication      | OTAA activation modes              |
| Firmware Protection | Secure boot and signed OTA updates |

## MacSync-WX1-BO Variants

| VARIANT NAME        | POWER SOURCE    | BATTERY CAPACITY | SOLAR SUPPORT | EXTERNAL FLASH | IDEAL FOR                                               |
|---------------------|-----------------|------------------|---------------|----------------|---------------------------------------------------------|
| MacSync-WX1-PO      | 12–24 V (Power) |                  |               |                | Industrial or commercial sites with stable power supply |
| MacSync-WX1-BO      | Battery         | 19 Ah            |               |                | Remote or portable deployments with low power           |
| MacSync-L-STD-SO-X1 | Solar + Battery | 12 Ah            |               |                | Outdoor, off-grid, and energy-autonomous environments   |

## What's in the Box?

Make sure box contains following items before installations



**MacSync**



**Mounting  
Clip**



**User manual &  
Warranty Card**



**Junction  
Box\***



**Mounting  
Screws**

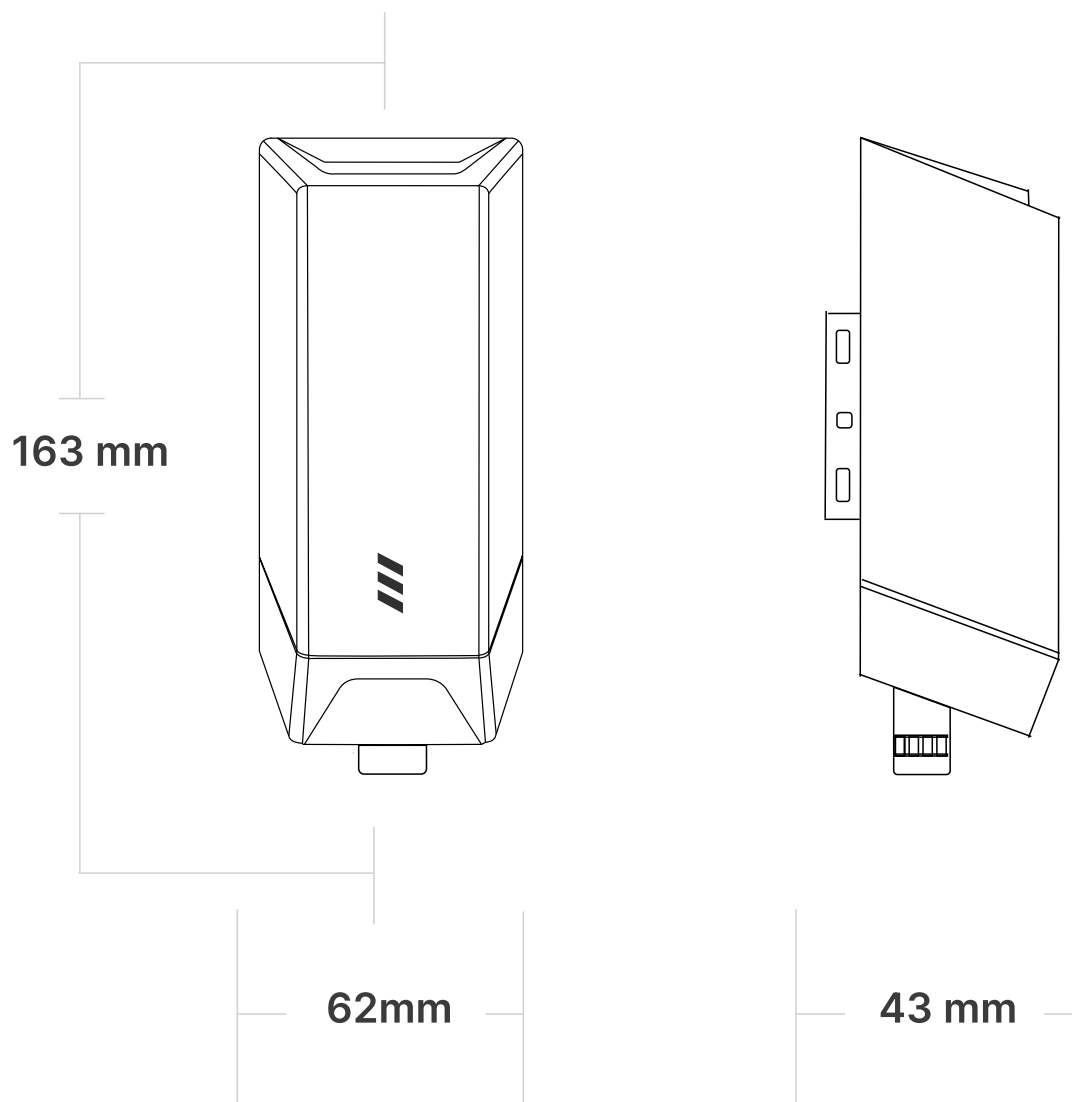


**Power  
Adapters\***

** Note**

Accessories must be purchased separately & may not be included in the box.

## MacSync-WX1-BO Dimensions



## Frequently Asked Questions

### What is the MacSync-WX1-BO used for?

MacSync-WX1-BO is a battery-operated RS485 and analog to Wi-Fi data logger for industrial and on-premise monitoring where wired power is unavailable. Typical applications include Industrial Automation & SCADA, Energy & Utility Metering, Process & Instrumentation Monitoring, Factories & Manufacturing Plants, Cold Storage & HVAC Systems and Building Automation & BMS.

## What inputs and outputs does the MacSync-WX1-BO have?

The MacSync-WX1-BO provides the following interfaces:

- RS485 Interface: 1 (Switchable)
- 4-20 mA Analog Channels: x 2 (Switchable)
- 0-10 V Analog Channels: x 2 (Switchable)
- Digital Inputs: x 2 (Switchable)
- ADC Resolution: 12-bit

## How does the MacSync-WX1-BO connect and send data over Wi-Fi?

The MacSync-WX1-BO connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band).

Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

## What is the IP rating and operating temperature of the MacSync-WX1-BO?

The MacSync-WX1-BO has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

## How is the MacSync-WX1-BO configured?

The MacSync-WX1-BO is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

## Which cloud platforms does the MacSync-WX1-BO work with?

The MacSync-WX1-BO can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

## Related Products and Resources

- User manual: [MacSync Wi-Fi datalogger user manual](#)
- Related datasheets: [MacSync-WX1-PO RS485 and analog to Wi-Fi data logger](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)